

Resolving the Hubble Tension: A Definitive NIR Calibration of the Cepheid Period-Luminosity-Metallicity Relation with JWST NIRCam

Abstract

The Hubble constant (H_0) is a fundamental parameter in modern cosmology, whose value determines the age, size, and expansion history of the Universe. Precise determination of H_0 is critical to constraining cosmological models, including the nature of dark matter, dark energy, and the curvature of the Universe. However, a significant 5σ tension has emerged between H_0 measurements derived from the early and late universe. The origin of this tension remains undetermined and may be attributable to systematics or requiring new physics beyond Λ CDM. We propose to obtain cutting-edge JWST NIRCam light curves of Gaia-calibrated Cepheids in the Milky Way and nearby galaxies to rigorously constrain the Cepheid Period-Luminosity-Metallicity (PLZ) relation. NIR photometry from NIRCam uniquely reduces dust extinction and metallicity sensitivity, yielding a first robust NIR PLZ calibration, superseding prior optical and UV datasets. This program will establish a new, precise measurement of H_0 and advance distance ladder cosmology by extending Cepheid calibration into an extinction-minimizing regime, directly addressing whether the Hubble tension is the product of systematic errors or a new signature of cosmological physics beyond the standard model. Beyond cosmology, a rigorously calibrated Cepheid PLZ relation will benefit all research areas that rely on precise, independent extragalactic distance measurements, such as stellar astrophysics and galaxy evolution. NIR photometry and spectroscopy of a large Cepheid sample will also constitute a lasting public legacy dataset.

■ Scientific Justification

Section 1: The Hubble Constant Tension and the Systematic Limits of the Distance Ladder

Precise determinations of the Hubble constant (H_0) reveal a statistically significant discrepancy between early- and late-Universe measurements, indicating either unaccounted systematics or physics beyond Λ CDM. Early-universe CMB observations yield $H_0 \approx 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ at $\sim 1\%$ precision under Λ CDM, while late-universe distance ladder measurements converge on $H_0 \approx 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$ at $\sim 1.5\%$ precision [4, 5]. This $\sim 5\sigma$ “Hubble tension” persists across independent probes and cross-checks. Resolving whether this discrepancy arises from residual systematics or new physics is therefore a central problem in precision cosmology, with direct implications for our understanding of dark energy, cosmic expansion history, and fundamental physics.

However, late-universe measurements of H_0 have entered a precision regime ($\sim 1\text{--}1.5\%$) where systematic uncertainties are comparable to statistical errors, critically limiting further progress. The dominant systematics arise from Cepheid calibration—dust extinction, metallicity dependence, and unresolved crowding—which can bias inferred distances at the $\sim 1\text{--}2 \text{ km s}^{-1} \text{ Mpc}^{-1}$ level [5]. Although insufficient to fully explain the $\sim 6 \text{ km s}^{-1} \text{ Mpc}^{-1}$ discrepancy, these biases are large enough to affect its interpretation and the inferred need for extensions to Λ CDM. Existing optical facilities have now reached their systematic floors in crowded extragalactic environments, where blending and reddening remain intractable. The inability to definitively characterize these systematics prevents a conclusive determination of whether the Hubble tension reflects new physics or residual measurement bias—making their resolution both urgent and essential.

We propose multi-epoch JWST NIRCам photometry and NIRSpec spectroscopy of Cepheid variables in the Milky Way—anchored by *Gaia* geometric parallaxes—and in nearby galaxies hosting Type Ia supernovae, to construct a robust near-infrared Period–Luminosity–Metallicity (PLZ) calibration. The James Webb Space Telescope is uniquely suited to this program: its NIRCам and NIRSpec capabilities operate at wavelengths intrinsically less sensitive to dust reddening, while its angular resolution mitigates crowding and blending that confound optical observations. Our strategy employs multi-epoch cadence observations to recover full Cepheid light curves across a broad period range, coupled with spectroscopic metallicity measurements to break PLZ degeneracies. These observations extend the Cepheid distance scale to a greater number of well-calibrated Type Ia supernova hosts, directly reducing the dominant systematics in the distance ladder and enabling a more precise, accurate determination of H_0 with enduring legacy value for precision cosmology.

Section 2: Unmodeled Distance Ladder Systematics Now Limit the Accuracy of Late-Universe H_0 Measurements

Late-universe determinations of H_0 using the Type Ia supernova distance ladder have reached $\lesssim 1.5\%$ precision, placing them in a regime where systematic uncertainties are comparable to statistical errors. However, several known but incompletely modeled systematics—dust extinction, metallicity dependence, unresolved crowding, and photometric calibration offsets—can bias H_0 at the level of $\sim 1\text{--}2 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [5]. While insufficient to fully resolve the $\approx 6 \text{ km s}^{-1} \text{ Mpc}^{-1}$ early-to-late tension, these biases are large enough to affect whether the discrepancy demands new physics, making a rigorous accounting of Cepheid systematics essential to advancing the field.

A critical missing component in current analyses is a homogeneous, high-resolution near-infrared (NIR) calibration of Cepheid variables spanning geometric anchors and supernova host environments. NIR observations suppress sensitivity to extinction and metallicity while reducing crowding-induced biases, enabling a more robust determination of the PLZ relation than is achievable with existing optical datasets. This program will provide precisely such a calibration by obtaining JWST NIRCам photometry and NIRSpec spectroscopy of Cepheids tied directly to the geometric anchors that define the distance scale, supplying the missing NIR calibration required to quantify and mitigate dominant systematics and enable a physically interpretable determination of H_0 .

Section 3: This Proposal: Multi-Epoch JWST NIRCам Photometry and NIRSpec Spectroscopy of Milky Way and Extragalactic Cepheids

We target calibrator Cepheids in the Milky Way and geometric anchor galaxies hosting Type Ia supernovae, obtaining time-series imaging and spectroscopy to refine the Cepheid distance scale at its foundational rungs. These sources directly set the zero point of the PLZ relation used in H_0 inference [5, 2]. JWST is essential to this program: NIRCам imaging and NIRSpec spectroscopy operate at wavelengths intrinsically less sensitive to dust reddening, suppressing extinction-driven systematics that dominate optical observations, while JWST’s angular resolution mitigates crowding and blending that limited prior *Hubble Space Telescope* campaigns in extragalactic fields [6].

We propose multi-epoch JWST NIRCам photometry and NIRSpec spectroscopy of Cepheid variables in the Milky Way and nearby galaxies with independent geometric distance anchors (parallax, water masers, eclipsing binaries). Observations employ a multi-epoch cadence sampling at least half the minimum pulsation period to recover full light curves and minimize aliasing systematics. NIRCам light curves will deliver high-precision mean magnitudes in multiple infrared bands, while NIRSpec spectroscopy provides direct metallicity constraints that break PLZ degeneracies. As illustrated in Figure 1 [adapted from 6], dispersion in the Period–Luminosity relation decreases systematically toward longer wavelengths, and JWST observations simultaneously reduce crowding-induced biases relative to *HST*—motivating our combined photometric and spectroscopic NIR strategy.

The primary product of this program is a robust, *Gaia*-anchored near-infrared PLZ calibration for Cepheid variables across Milky Way and extragalactic environments. This calibration directly reduces the dominant systematic uncertainties—extinction, metallicity dependence, and crowding—enabling a more accurate and reliable determination of H_0 .

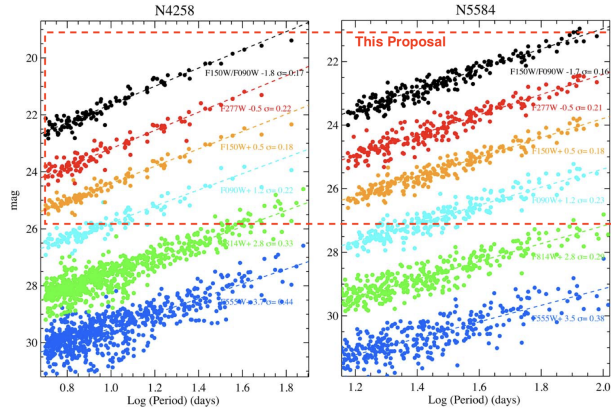


Figure 1: JWST NIRC*am* observations enable a systematic-limited calibration of the Cepheid distance scale. Plotted are Period–Luminosity relations spanning bluer *HST* bands (lower) to longer-wavelength JWST NIRC*am* bands (upper); the uppermost relation (black) shows a Wesenheit magnitude constructed from F150W and F090W. Dispersion decreases at longer wavelengths, and JWST’s angular resolution reduces crowding-induced scatter relative to *HST*. Current calibrations are limited by extinction, metallicity sensitivity, and blending that JWST uniquely mitigates, reducing systematics that propagate into H_0 . [Figure adapted from [6]].

Section 4: A Local and Extragalactic Legacy Dataset of NIR Photometry and Spectra for Cepheid Variable Stars

The proposed NIR PLZ calibration will minimize the systematic effects of extinction, blending, and metallicity that currently limit Cepheid-based distance measurements, directly improving the reliability of H_0 inference from the distance ladder. These products will advance the field by determining whether the Hubble tension requires new physics beyond Λ CDM or is attributable to residual observational systematics [5, 1].

This program is timely because JWST is now operational, enabling multi-epoch Cepheid datasets that extend beyond the systematic limits of existing optical surveys and expand the number of calibratable Type Ia supernova host galaxies. It is well aligned with JWST’s core science goals—probing the expansion history of the Universe, tracing galaxy evolution, and constraining stellar life cycles—delivering precision constraints on H_0 , mapping stellar populations in host galaxies, and refining pulsation models for intermediate-mass stars. This study will increase the long-term legacy value of JWST by producing a publicly accessible, high-precision NIR Cepheid photometry and spectroscopy dataset supporting stellar structure, galactic chemical evolution, and cosmological distance scale research—establishing JWST as the definitive facility for distance ladder cosmology.

Bibliography

References

- [1] Cai, R.-G., et al. 2022, *Phys. Rev. D*, 105, 103513
- [2] Freedman, W. L., et al. 2019, *ApJ*, 882, 34
- [3] Lindegren, L., et al. 2021, *A&A*, 649, A2
- [4] Planck Collaboration 2020, *A&A*, 641, A6
- [5] Riess, A. G., et al. 2022, *ApJ*, 934, L7
- [6] Riess, A. G., et al. 2023, *ApJL*, 956, L18